Early Signals of Student Dropout

Finding the Line Between Detection and False Alarms

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Executive Summary

Universities often identify students who are struggling only after serious academic problems have emerged. Could information already collected during the first semester help institutions recognize students at risk of dropping out? More importantly, would those predictions be sufficiently useful to justify an intervention?

This study examined 4,424 student records from the UCI *Predict Students' Dropout and Academic Success* dataset. Three first-semester indicators were analyzed: approved curricular units, average grades, and completed evaluations. The differences were substantial. Among students with no approved curricular units, 79.4% eventually dropped out, compared with just 10.0% of students who completed six units.

A logistic regression model was developed using 3,630 students with known graduation or dropout outcomes. At a 50% classification threshold, the model achieved 83.1% accuracy and correctly identified 74.3% of students who eventually dropped out. Lowering the threshold to 40% increased dropout detection to 79.9%, with accuracy declining slightly to 82.6%. In the 726-student test sample, this change identified 16 additional dropouts but also generated 19 additional false alarms.

These results illustrate an important institutional decision. Lower thresholds identify more students who may need assistance, but they also increase the number of students requiring follow-up. Whether that tradeoff is worthwhile depends on the cost, effectiveness, and design of the intervention. A brief initial check-in may justify a different threshold than a resource-intensive advising appointment.

The findings support further investigation through a limited pilot program rather than immediate institution-wide implementation. Such a pilot should validate the classification threshold, assess advising workload, investigate students with unusually low academic completion, and monitor differences in model performance across student groups. Most importantly, it should evaluate whether the resulting interventions actually improve student retention.

The central finding is that predicting student dropout and preventing it are two different problems. First-semester academic records provide useful warning signals, but the value of an early-warning system ultimately depends on whether universities can translate those predictions into effective student support.

1. The Decision Behind the Research Question

The question this study seeks to answer is intentionally narrow: How well does first-semester academic performance indicate individuals at high risk of dropping out of university? The proposed

warning system is based on completed first-semester performance, not necessarily a system that operates during the first semester. The availability of the information is the key to whether or not an institution should truly consider employing data in an early-warning program.

I was also looking for limitations to the predictive value of the academic record itself. First-semester course outcomes are part of the normal flow of the university system and are therefore more accessible to an academic-support office than a new course design, psychology evaluation tool, or intricate student-profile system. My interest is not simply in developing another predictive model, but in determining whether existing institutional data can be used to improve student outcomes.

What Counts as a Useful Prediction?

A model may have good prediction accuracy, but still fail as an advising tool. If it misses students who leave, it foregoes an opportunity to help those students. If it marks an excessively high number of successful students as being in danger, the university is probably wasting staff time following up on warnings that do not warrant rigorous attention. The concern is what happens when the university acts on an incorrect prediction, particularly in an office with a limited number of students it can assist and appointment times available.

Mistakes (and therefore their associated costs, perhaps asymmetrical) come in two types: students who later drop out and were not flagged (false negatives), and students who graduate but had received a flag (false positives). While a simple email of encouragement might be harmless, if it meant additional meetings and appointment times, the university could quickly run out of appointment slots and employees. The meeting itself might also make a student feel labeled as below average. Clearly how the intervention is designed would be just as important as where to place the probability threshold.

Scope of the Project

I used actual student records from previous years to see whether this idea might work. That does not mean I can predict what will happen to every student, and it certainly does not mean the university could prevent every dropout. What I was trying to determine was whether the results were promising enough for the university's board to consider a small pilot program. Would it be worth having staff spend some time testing this approach before making a larger commitment?

2. Data, Preparation, and Workflow

For this project I used Predict Students' Dropout and Academic Success, a dataset from the UCI Machine Learning Repository (Realinho et al., 2021). It has 4,424 student records with 37 variables. The students fall into three groups: 2,209 graduated, 1,421 dropped out and another 794 were still enrolled. There were no missing values in the data.

I opened the CSV file in pandas, although I had to account for the semicolons separating the values. I checked the first few rows, cleaned up the column names and looked for missing values or problems with the data types. For the actual analysis I decided to concentrate on just three things from the first semester: how many curricular units students passed, their average grade and how

many evaluations they had. The variables used in the analysis are described in Appendix B. I could have used more variables, but I wanted a model that would be fairly simple to explain and that somebody else could reproduce without too much trouble. I also wanted to see how much I could learn from academic performance alone, without bringing in personal or demographic information about the students.

A Consequential Modeling Decision

Students that were described as Enrolled had to be excluded because it was not possible for me to determine their eventual graduation or dropout status from these records. This means that only a sub-sample of 3,630 students was considered for the model and the results of this work relate strictly to them. I am not sure whether the students who are still enrolled are similar enough to those who graduated or dropped out for the model to work equally well with them.

The students were allocated between the 2,904 case train and 726 case test groups, stratified by outcome and based on a fixed random seed. Logistic regression was then applied where the 3 numerical predictors were standardized by a scikit-learn pipeline. The fitted model generated dropout probabilities and binary predictions for the test group. Accuracy, precision, recall and a confusion matrix were reported afterwards. The effect of using probability thresholds of 50%, 40%, 30% and 20% was investigated to illustrate what happened if the size of the group flagged changed, based on whether the university was more or less willing to overlook a potential dropout. A summary of the data preparation and modeling workflow is provided in Appendix C. All calculations, model results and figures presented in this report were generated using my Python analysis notebook. My use of AI assistance in developing and debugging the code is described in Appendix A. The staffing and retention examples discussed later are hypothetical and were not calculated from actual intervention data.

What Was Not Done

No missing values were imputed (as there were none in the dataset). Students with zero grades were retained in the analysis. Although these values warrant further investigation, excluding them without understanding how they were recorded could introduce bias. I also avoided unnecessarily complicated methods or adding dozens of demographic variables just to improve my model results. These were self-imposed boundaries and not intended to highlight an alleged absence of measurement problems in the dataset. There is still a question about the first-semester variables. Before using them with actual students, I would want to know exactly when each was recorded and whether the university defines them the same way as the original dataset.

3. First-Semester Performance: A Visible Separation

I was expecting these differences, but the extent of the contrasts was worth noting. Students who finally graduated averaged 6.23 approved first semester units, vs 4.32 for those still enrolled and 2.55 for those who dropped out. The average grades followed the same pattern: 12.64, 11.13 and 7.26, respectively. They are group differences, not predictions about individual students, but they show that first-semester performance may be valuable in identifying learners at risk.

Figure 1

Mean First-Semester Approved Units by Student Outcome

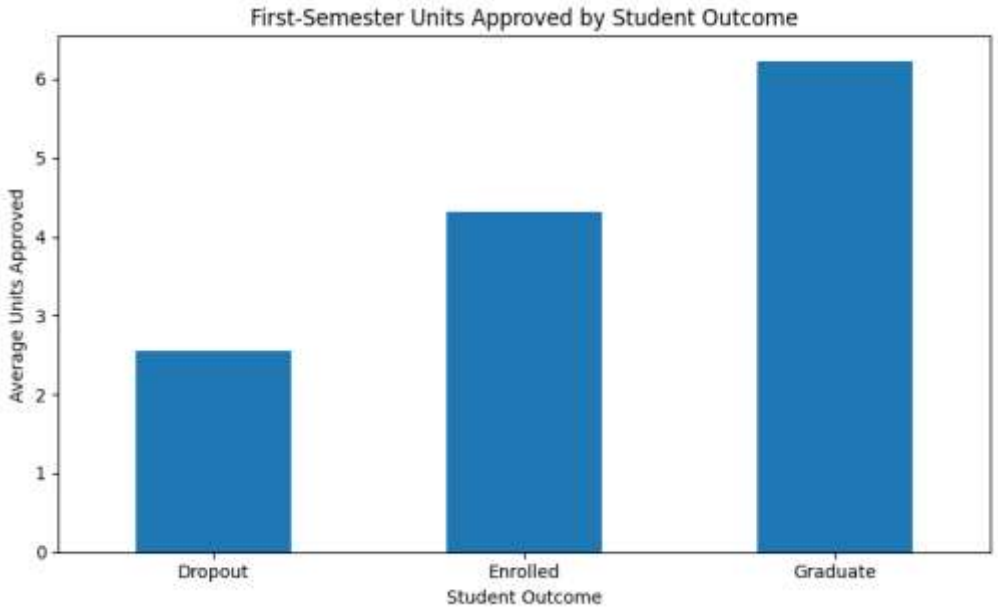
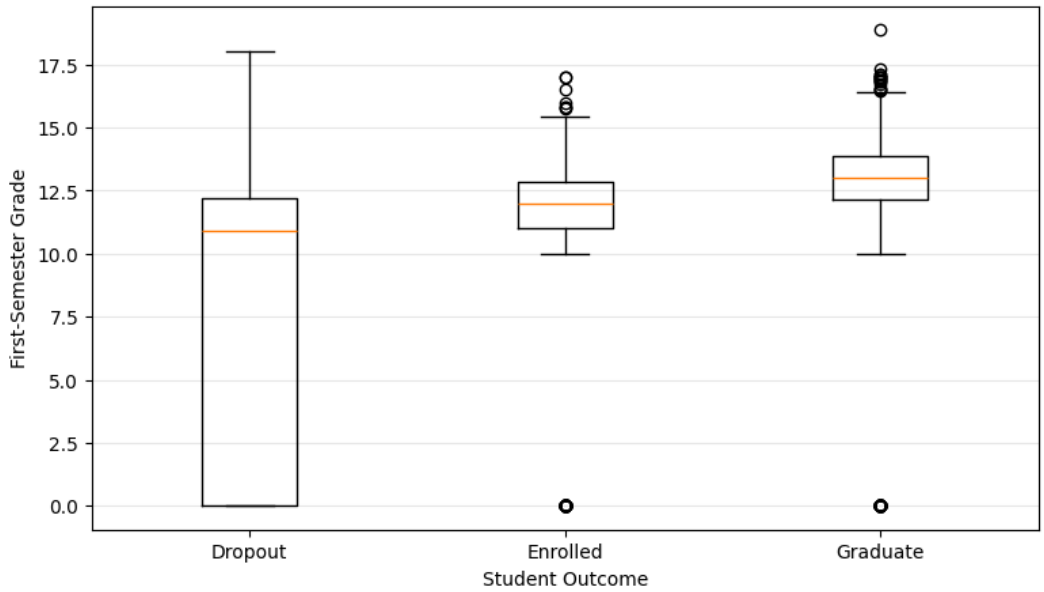


Figure 2

Boxplots of First-Semester Grades by Student Outcome



Note. First-semester grades are measured on a 0–20 scale, with higher values indicating better academic performance.

The difference in approved units is substantial. Figure 1 shows just how different the groups were when it came to approved units. It does not take an advanced algorithm for a university to notice that students passing very few units may be in trouble. But when should the university actually do something about it? And what should they do? That is more difficult to answer. Having a rule based only on approved units might allow me to identify many students who are in difficulty, but it might leave aside those who, despite passing most of their units, are still in trouble.

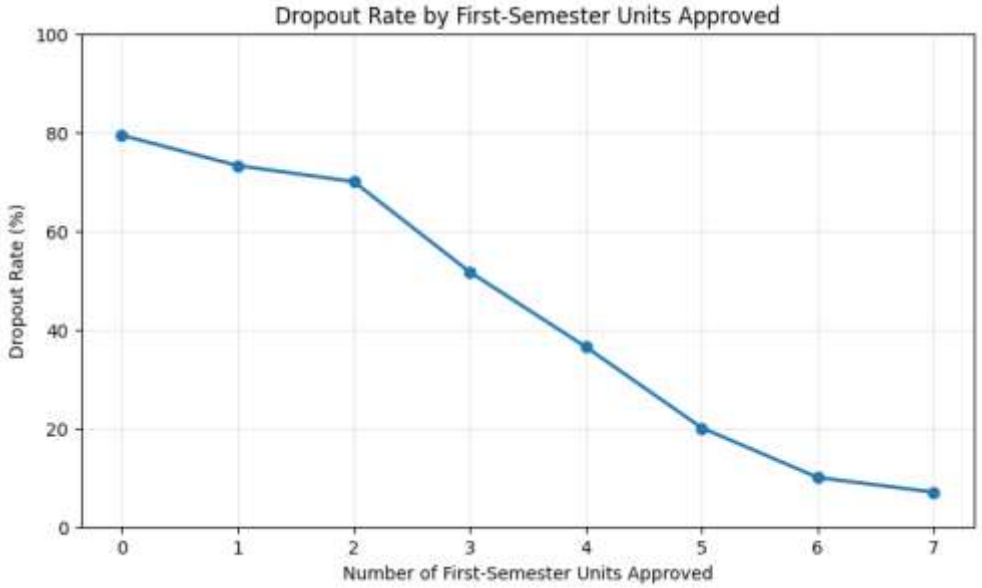
Another source of trouble comes from looking at the distribution of first-semester grades. As the three boxplots in Figure 2 show, the grades overlap quite a bit between the three groups. The dataset contained numerous zero grades, although their meaning was unclear. Some may reflect missed examinations or incomplete coursework. Importantly, several students who eventually dropped out had relatively strong grades, while some graduates had performed poorly. The substantial overlap makes it difficult to establish a reliable cutoff using grades alone. This was one of the reasons for choosing a probabilistic model, which can take into account more than one measure.

4. Approved Units and the Dropout Gradient

There was something worth a second look in the averages. I divided the students into groups based on how many first-semester units they had approved. Of particular relevance was the group with none: 718 learners of whom 570 dropped out (79.4%). Conversely, just 10.0% of students with six approved units were reported as dropouts. Much of the fall in dropout rates happened between two and six approved units. Figure 3 shows how the dropout rate changed as the number of approved units increased.

Figure 3

Observed Dropout Percentage by Approved First-Semester Units (Zero Through Seven)



The difficulty with these statistics is their interpretation. Would helping a student pass another curricular unit actually reduce their likelihood of dropping out? This dataset cannot answer that question. Students already considering leaving may put less effort into their courses, meaning poor academic performance could be a consequence of their decision rather than its cause. Academic difficulties could also contribute to the decision to leave.

Finding a student with almost no approved units is one thing. Knowing what to do about it is another. Before deciding whether the institution can help, I would want to know why the student made so little progress. The problem may have nothing to do with something tutoring could fix. The graph identifies students the university might want to contact (not necessarily what is wrong, or if the university can help).

5. Anomalies, Correlations, and Boundaries of Simple Rules

The boxplots in Figure 2 revealed an unusually large number of zero grades. There were 718 students with a first-semester grade of 0, all of whom also had 0 units approved. Of those students, 570 dropped out, 77 graduated and 71 were still enrolled. I was not sure what to make of those zero grades without knowing how the university records them. I decided to leave these students in the study, but a larger study would need to investigate what those zeros meant.

A small number of students also completed an unusually large number of curricular units. The numbers of students in these groups are so small that even one or two cases can make a large impact in the proportion of dropouts. Therefore Figure 3 displays only students with zero to seven approved units. There were still students with higher numbers in the model.

Correlations Among the Predictors

Approved units and grades are fairly strongly related, which makes sense because students who pass more curricular units would generally be expected to have better grades. However, the relation clearly is imperfect and more information might be contained in the "evaluations" variable. I retained all three predictors because each may contribute information not fully captured by the others, despite the correlations among them.

Table 1

Pearson Correlations Among First-Semester Predictors

Pair of first-semester measures	Pearson r
Approved units and grade	0.696
Approved units and evaluations	0.522
Grade and evaluations	0.418

A practical limitation is that the number of approved curricular units may not be available until the semester ends. If the goal is to help students avoid dropping out early, it may already be too late to learn about their academic performance when the semester is over. Maybe midterm grades or the number of evaluations a student completed could give universities some hints, allowing them to intervene earlier, but I did not test that possibility.

6. How the Model Performed

The model was trained on 2,904 students and tested on another 726. The model correctly predicted the outcome for 83.1% of students in the test group. For dropouts, precision was about 81% and recall was 74.3%. Recall here means how many of the students who actually dropped out were identified by the model, using a probability threshold of 0.5. Graduates accounted for 60.9% of the test group, so even though the model made mistakes, it did considerably better than predicting that every student would graduate.

Table 2

Confusion Matrix for Dropout Predictions at the 50% Probability Threshold

Actual outcome	Predicted graduate	Predicted dropout
Graduate	392	50
Dropout	73	211

Of the 284 students who dropped out, 211 were correctly predicted by my model with the threshold at 0.5. The other 73 were students who dropped out but were not identified by the model, or in other words, false negatives. The 50 students wrongly identified by my model as dropouts did not drop out at all; they graduated. These were the false positives. The 73 dropouts I could not identify represent students who might have benefited from additional assistance. Either way, the university could end up spending time responding to false alarms.

Why Not Simply Maximize Recall?

I could lower the bar at which a student is flagged and likely identify more students who end up dropping out. The practical question is what consequences follow from issuing these warnings. If the university expects an advisor to arrange a one-to-one meeting with each student, it could mean a lot of extra work, especially when some of those students would have graduated anyway. However, if warning a student only means offering a few extra tutoring sessions or just checking to see if they need assistance, then the costs of a few more false positives might not be that high.

These performance measurements are backward-looking and based on students with known outcomes, so I should not assume the same results will hold for the next entering class. My exploration of other thresholds is preliminary and would need separate validation.

7. The Threshold Is an Institutional Choice

Lowering the classification threshold increases the number of students correctly identified as future dropouts, but also increases the number of graduates incorrectly flagged as at risk. Figure 4 shows what happened to both numbers as I lowered the threshold.

Figure 4

Detected Dropouts and False-Positive Alerts at Four Probability Thresholds (726 Test Students)

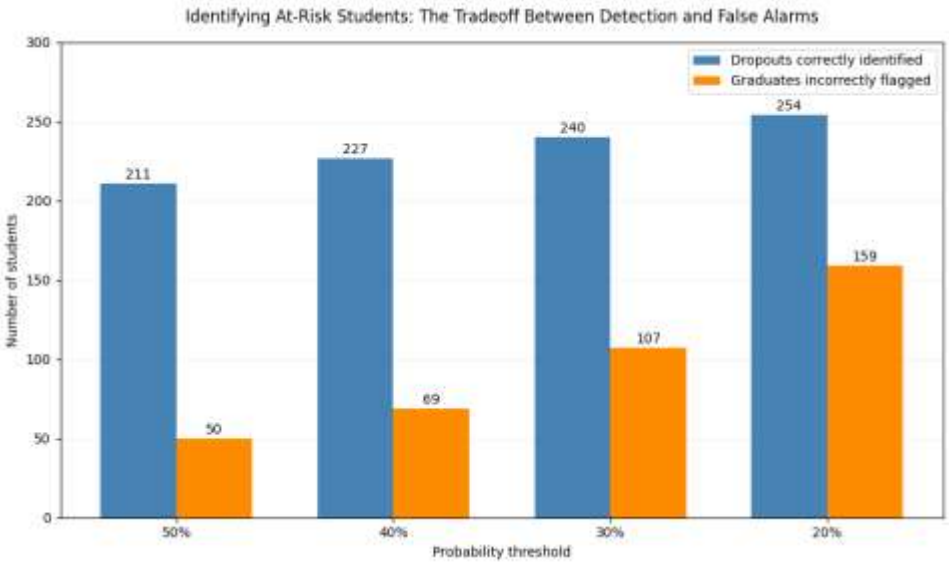


Table 3

Dropout Detection and False Positives at Four Probability Thresholds

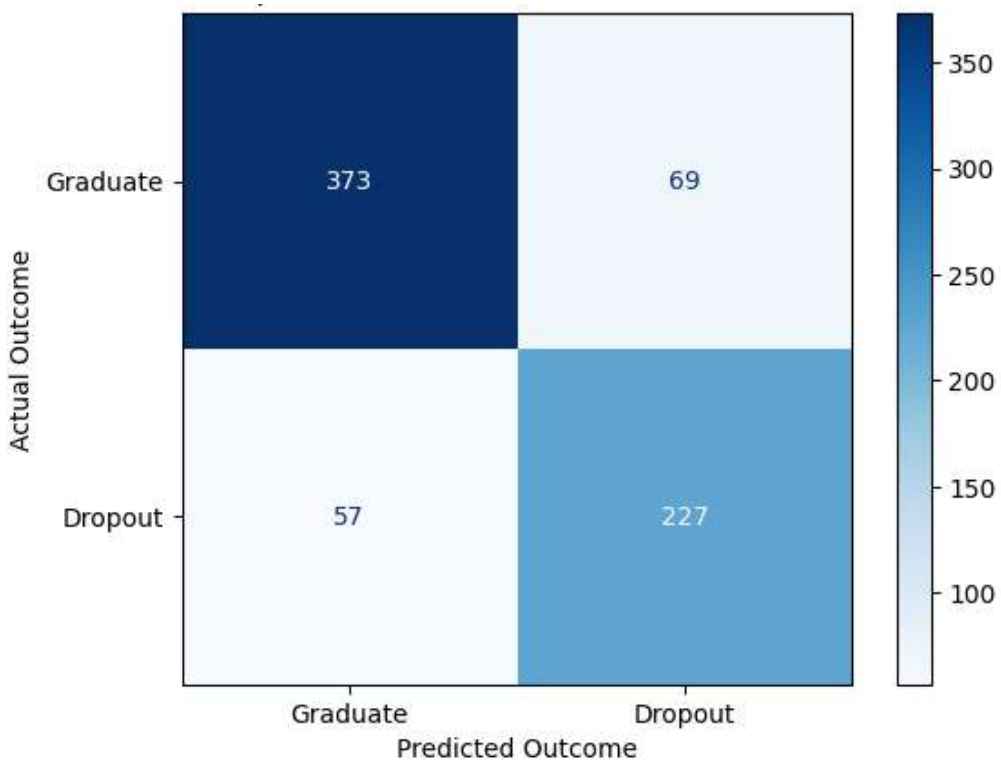
Cutoff	Detected dropouts	Missed dropouts	False positives
50%	211	73	50
40%	227	57	69
30%	240	44	107
20%	254	30	159

Lowering from 50% to 40% correctly identifies another 16 students who drop out whereas 19 new false positives are added. Reducing from 40% to 20% identifies another 27 but at the cost of 90 new false positives. Thus, the last change is much more questionable if each flag prompts an expensive intervention. It could however be justifiable if that intervention is cheap or even good for students regardless of whether or not they drop out.

For this project, I chose 40% as an experimental threshold. This led to a recall of 79.9%, precision of 76.7% and accuracy of 82.6%. Figure 5 shows the confusion matrix at this threshold, including the 227 dropouts correctly identified, 57 missed dropouts and 69 false alarms. That said, this does not imply I have demonstrated that 40% is the optimal threshold.

Figure 5

Confusion Matrix for Dropout Predictions at the 40% Probability Threshold



Because I compared the thresholds using the test data before choosing 40%, its reported performance is no longer a fully independent evaluation. A better way would be to compare thresholds on training and validation data, and make the selection there, and only then analyze it on a separate test set that was not used to make the choice. I did not go through the complete procedure here, so the 40% threshold is a tentative option, not a verified recommendation.

8. Recommendations: From a Score to a Support Process

Recommendation 1: Pilot a 40% Alert Threshold

I recommend initially testing the 40% threshold with a single entering cohort or a limited number of academic programs. This does not indicate that I have identified the best potential cutoff, but that I am starting with the result from this study. So, I would have to do further testing to see if 40% is really a reasonable decision.

The 40% threshold resulted in 296 alerts, out of which 227 were true positives and 69 were false positives for the 726 students in the test sample. This information helps me estimate the advising workload, keeping in mind that there would of course be variations in real life. The test only included students with known outcomes, whereas an actual university would be dealing with students

whose eventual outcomes are not yet known. The success and dropout rates could also be different from those in the sample.

A pilot producing the most alerts would not necessarily be the most successful one. I am especially interested in whether similar results can be repeated and whether the university has the capacity to deal with this volume of alerts.

Recommendation 2: Separate Low-Cost Contact from Intensive Advising

An alert should lead to a respectful check-in with a student, not some kind of label being attached to them. An initial check-in could establish whether the student needs assistance. Students requiring additional support could then be referred for a more comprehensive advising appointment. Otherwise, there may be no reason to take things any further.

For planning purposes, assume each initial student contact requires 10 minutes and a more comprehensive advising appointment requires 30 minutes. Contacting 300 students would require approximately 50 staff hours. If 25% subsequently needed additional advising, those 75 appointments would require another 37.5 hours, bringing the estimated total to 87.5 staff hours.

I have not timed any real interventions and the estimate does not cover scheduling, paperwork or other administrative labor. The purpose is to estimate the staffing commitment before deciding how many students to flag.

Recommendation 3: Give Very Low Completion a Prompt Human Review

The dropout rates were 79.4% for students with zero approved units, 73.2% for students with one and 70% for students with two. However, I would be hesitant about presuming that fewer units passed was the cause of these students dropping out. What the figures tell me is that students with very low completion rates are worth investigating.

Before doing anything else, staff definitely should look into why the student passed so few units. Did they leave any courses? Did they adjust the load of courses? Or is part of this attributable to how the institution tracks academic progress? There is also the question of whether the student is encountering problems that someone at the institution might truly help with. I cannot answer that merely by looking at how many units they passed.

I would suggest reviewing and documenting these cases within 5 business days of an alert. This would allow me to see what proportion of those highlighted instances really received that review. Research on early-alert systems suggests that identifying students who are struggling can provide useful information for intervention efforts (Villano et al., 2018). Whether responding within five business days actually improves retention is something the pilot would need to investigate. It is one matter to have a computer identify learners who may be struggling, while it is another for the school to follow up and see whether they need help.

9. Recommendations Continued: Measure What Happens After Contact

Recommendation 4: Evaluate Intervention Outcomes (Not Only Model Scores)

Having a classifier that works may be a start. However, it does not mean the university has a retention program that works. Bettinger and Baker (2014) found that individualized student coaching improved college persistence in a randomized experiment. Proper evaluation requires keeping track of the actual students: who got contacted, what sort of help was offered, how many took it up, and what happened to them the next semester. I am also not sure what kind of help is worth offering.

Students could be randomly offered different kinds of assistance and then followed to see what happens. A university could also try comparing the students who got help against others who were in roughly the same academic situation but did not receive help. Again, that is not a clean comparison, though, for the students who accept help may already be different (e.g., some of them may have already reached out to the university themselves), and you cannot immediately attribute any improvement to the help provided.

Say the university worked with 200 students. If 90% of them would have come back anyway and that figure rose to 95%, you would have 10 more students returning the next semester. Whether it is worth it possibly depends on the cost of the staff time spent helping those students. More importantly, I would want to be certain the students stayed because of what the university did and not because of other reasons. It seems like a reasonable thing to aim for, but nothing in the present analysis shows the university could actually achieve it. That would have to be tested separately.

Recommendation 5: Establish a Recurring Model Review

The assumption that this model makes about the same number of mistakes across academic programs might not be true; it may be relatively competent with one major while being much worse with another. The results should be examined separately and, again, the matter of when the information became available needs to be addressed. The model may identify students who are likely to drop out, but will it be able to perform this task early enough so that some action can be taken on that information?

Universities also change. Curricula get revised, and the mix of students can shift. For instance, more students might begin arriving from rural areas. I would not just assume that the model would continue doing as well as it did before. It should be evaluated at least once per semester, and a determination would have to be made as to whether it needed updating.

A proper evaluation would look at the students the model missed, the staff time spent dealing with the alerts and whether anyone did not want to be contacted. The 40% threshold could be worth reviewing as well. It might be manageable. Or it might create more work than anticipated. The model should not operate indefinitely without oversight. The university should designate responsibility for monitoring its performance, reviewing alerts, and evaluating intervention outcomes.

Python can report how many dropouts it correctly identified. What the university actually needs to know is how many students it managed to keep enrolled.

10. Risks, Implementation, and Next Steps

Risks Requiring Explicit Mitigation

There are several things I would want to check before the university started using this model. Appendix D summarizes the issues that would need to be addressed before implementation. Firstly, I would check when the data actually made its way into the university's system. By the time final grades for the first semester are available, it may be too late to start helping the neediest students. Would information available earlier in the semester support sufficiently reliable predictions?

This also applies to students who are yet to graduate. I have excluded these students, but still do not know whether they will graduate or drop out. It may be possible that the model works differently for the present group of students. It would be best to watch them at intervals over time.

I would also want to find out whether or not the model makes more mistakes for students in some groups or academic programs. Another measure would be to let students question a warning that they think was issued by mistake. There are the strange values in the dataset. Some students have a grade of 0, while others have an unusual number of approved units. What these values stand for is something I want to know before I can count on them.

Last would be the issue of privacy. These predictions should not be available to everyone at the university. I would also want some boundaries on who gets to see these predictions and what they are allowed to do with them. An at-risk designation should be treated as a prediction, not as evidence of student misconduct or failure.

Proposed Rollout

Weeks one and two would probably be spent learning exactly what information the university has available, what the individual fields refer to and when this information becomes available in the academic year. During weeks three and four I would again try performing my analysis using the university's own historical records. I would also want separate groups of students for training, validation and final testing. In weeks five and six I would want to talk to the academic advisors. How many students can they realistically contact, and how can students be approached in such a way that they do not feel singled out as likely failures?

Next, I would like to trial the system on a small scale for one term, with careful attention paid to students contacted (what happened next, how much work was involved, etc.). At the end of the semester, I would know how many valid predictions I had made, how many false alarms there had been, and how many students the model had missed. I also want to evaluate if the strategy has a different effect for different groups of learners and if the assistance is of any benefit at all.

Before rolling out the program, it is worth investigating if the data helps in making reasonable predictions. Also, do these predictions lead to overloading the advisors? Finally, are the interventions actually helping student retention?

What the Project Establishes (And What it Does Not)

The evidence supports a modest but worthwhile conclusion. Three first-semester performance measures provide useful information about whether students from this dataset drop out or graduate. Much of that can be seen using simple logistic regression, and changing the threshold helped me consider the tradeoff between identifying more students and creating additional work for university staff. Still unknown is whether the model would behave the same way in a new school, whether its inputs would be available early enough in the term to help, or whether acting upon the predictions would improve student retention. The next step should not be about adding more algorithms, but about determining whether the university can use these predictions to help students remain enrolled.

References

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Villano, R., Harrison, S., Lynch, G., & Chen, G. (2018). Linking early alert systems and student retention: A survival analysis approach. *Higher Education*, 76(5), 903–920. https://doi.org/10.1007/s10734-018-0249-y

Appendix A

AI Use Disclosure

ChatGPT was used to assist with Python programming and debugging, particularly when implementing unfamiliar commands or resolving errors. I reviewed the resulting code and checked the statistical outputs for consistency with the intended analyses. The research question, analytical decisions, interpretation of findings, and recommendations were my own. AI-generated suggestions were evaluated rather than accepted without verification.

Appendix B

Data Dictionary

The original dataset contains 37 variables, but this analysis used three predictors and one outcome variable.

Variable	Description	Type / Role
Curricular units 1st sem (approved)	Number of first-semester curricular units successfully completed	Int/Predictor
Curricular units 1st sem (grade)	Average first-semester grade, measured on a 0–20 scale	Num/Predictor
Curricular units 1st sem (evaluations)	Number of first-semester curricular unit evaluations	Int/Predictor
Target	Student outcome: Dropout, Enrolled, or Graduate	Cat/Outcome

Note. The definition of variables was based on Realinho et al. (2021). Students designated as Enrolled were not included in the logistic regression as their ultimate results were uncertain. The model differentiated Dropout from Graduate. Int = integer; Num = numeric; Cat = categorical.

Appendix C

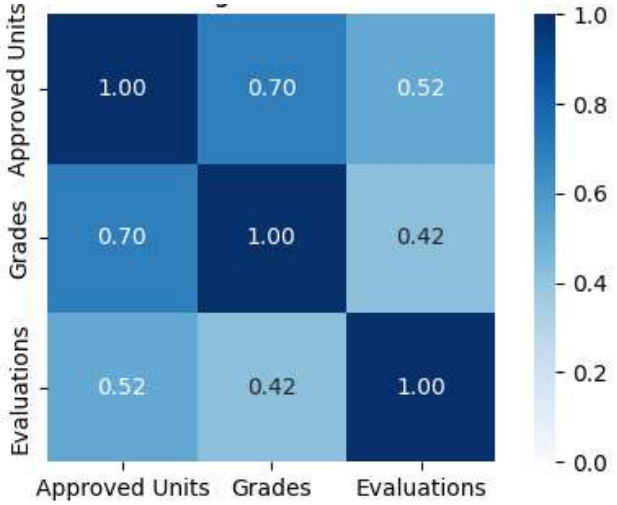
Data Preparation and Modeling Workflow

Step	What was done
Data import	Loaded the UCI dataset into pandas, using “;” as delimiters.
Initial inspection	Checked the dataset's dimensions, column names, data types, and missing values.
Variable selection	Selected three first-semester academic predictors.
Outcome preparation	Excluded 794 Enrolled students, which leaves 3,630 records that have known outcomes.
Data partitioning	Created a stratified training set of 2,904 students and a test set of 726.
Model fitting	Used scikit-learn to standardize predictors and fit a logistic regression model.
Evaluation	Examined accuracy, precision, recall, and confusion matrices.
Threshold analysis	Compared probability thresholds of 50%, 40%, 30% and 20%.

I started by looking at the number of approved units for students with different outcomes (Figure 1), followed by their first-semester grades (Figure 2). Both helped me decide which predictors to use in the model. I also created a correlation heatmap (Figure C1) to look at the relationships among approved units, grades, and evaluations.

Figure C1

Correlations Among First-Semester Predictors



Note. Values used are Pearson correlations among three academic indicators in the first semester. The strongest association is between approved units and grades (r = .70), followed by approved units and evaluations (r = .52) and grades and evaluations (r = .42).

Appendix D

Model Limitations and Validation Requirements

Issue	What needs to be checked before implementation
Timing of data	Check to see if academic information is available early enough to lead to helpful assistance.
Zero grades	Establish what zero grades and zero approved units actually represent in institutional records.
Generalizability	Test whether the model performs similarly with new students and at other institutions.
Threshold selection	Select and validate the alert threshold using separate validation and test data.
Student differences	Look for differences in prediction errors by academic program and student group.
Intervention effectiveness	Determine whether contacting flagged students actually improves retention.